PROJECT MANAGEMENT 1

Monday 13th October 2025

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Project Management - Learning objectives

Unit level learning objective(s)

2. work as a member of a team, employing appropriate project management and planning tools to create, monitor and deliver a project plan;

Session level learning objectives

1. **Recall** and **discuss** the attributes of waterfall and agile project management approaches
2. **Create** a work breakdown structure
3. **Identify** task dependencies within a project
4. **Create** a network diagram from a list of task durations and task dependencies
5. **Create** a Gantt chart from a corresponding network diagram (or directly from a list of tasks and task dependencies) (NEXT WEEK)
6. **Identify** the critical path of a schedule and the float of any task within a network diagram or Gantt chart
7. **Critically** analyse a project schedule to **identify** risks and opportunities within the plan (NEXT WEEK)

What is a **Project**?

- **Project:** A planned set of interrelated tasks to be executed over a **fixed period** and within certain cost and other **limitations**
  - Temporary. Defined beginning and end. Fixed scope. Fixed resources. Unique (not routine).

- **Programme:** multiple projects to achieve overall objectives,
  - Common knowledge and skills across projects.

- **Portfolio:** multiple projects & programmes to achieve **strategic direction** and objectives

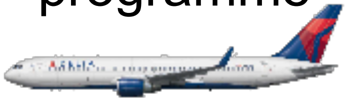
Examples from Aerospace

Portfolio: Widebody Jets

747 programme

- Project 1
- Project 2
- Project …

767 programme



- Project 1
- Project 2
- Project …

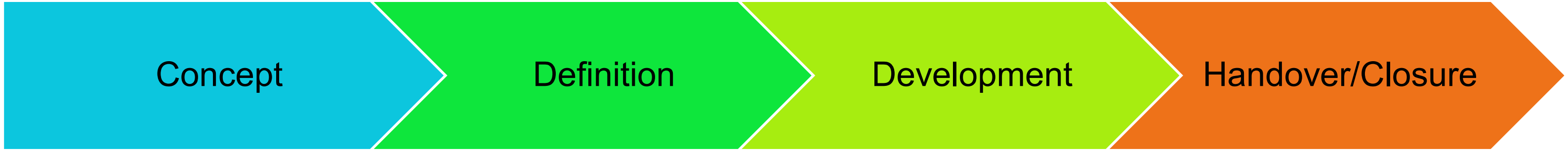
777 programme

- 777-300ER
- 777F
- 777-9X
- 777-8X

787 programme

- Project 1
- Project 2
- Project …

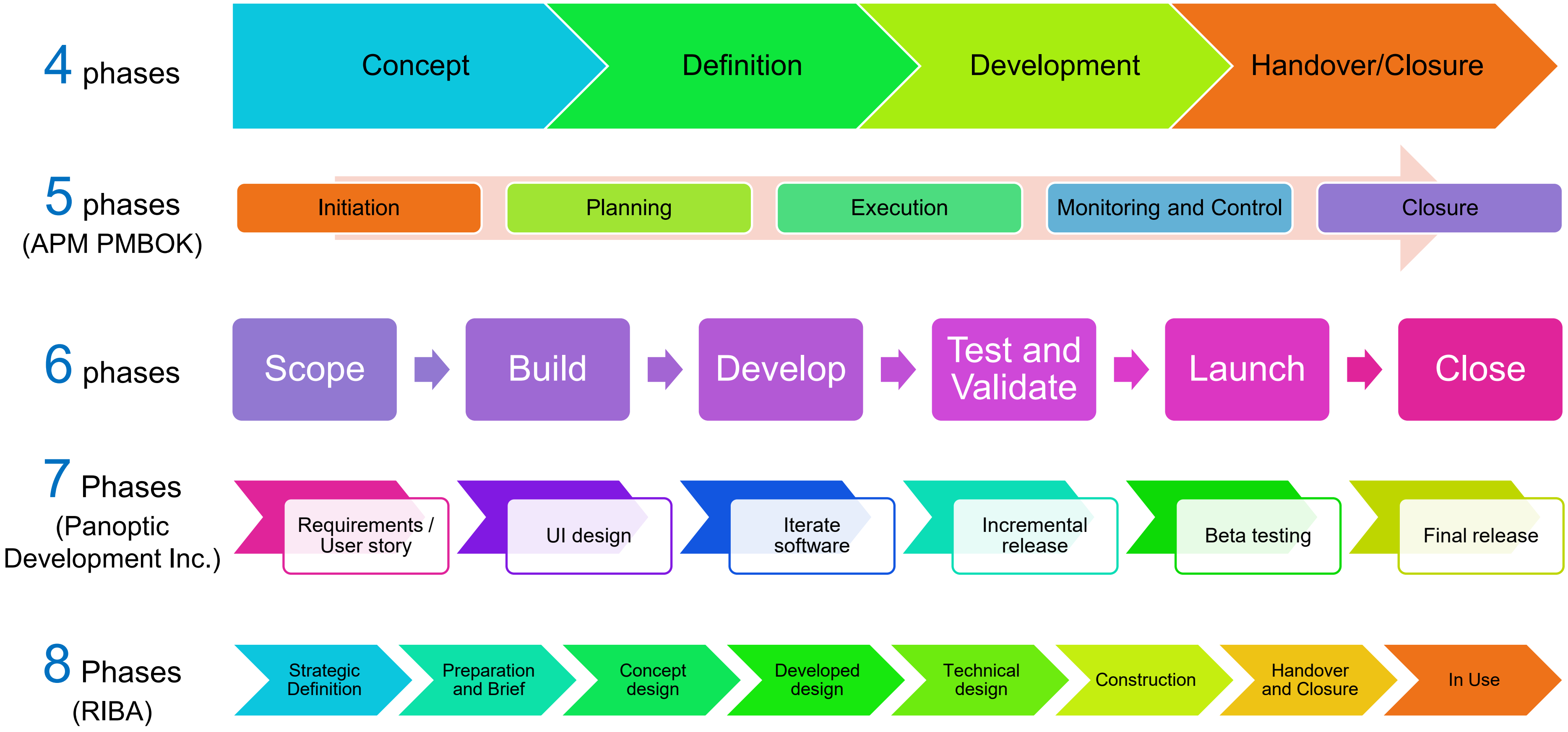
Project Lifecycle(s)



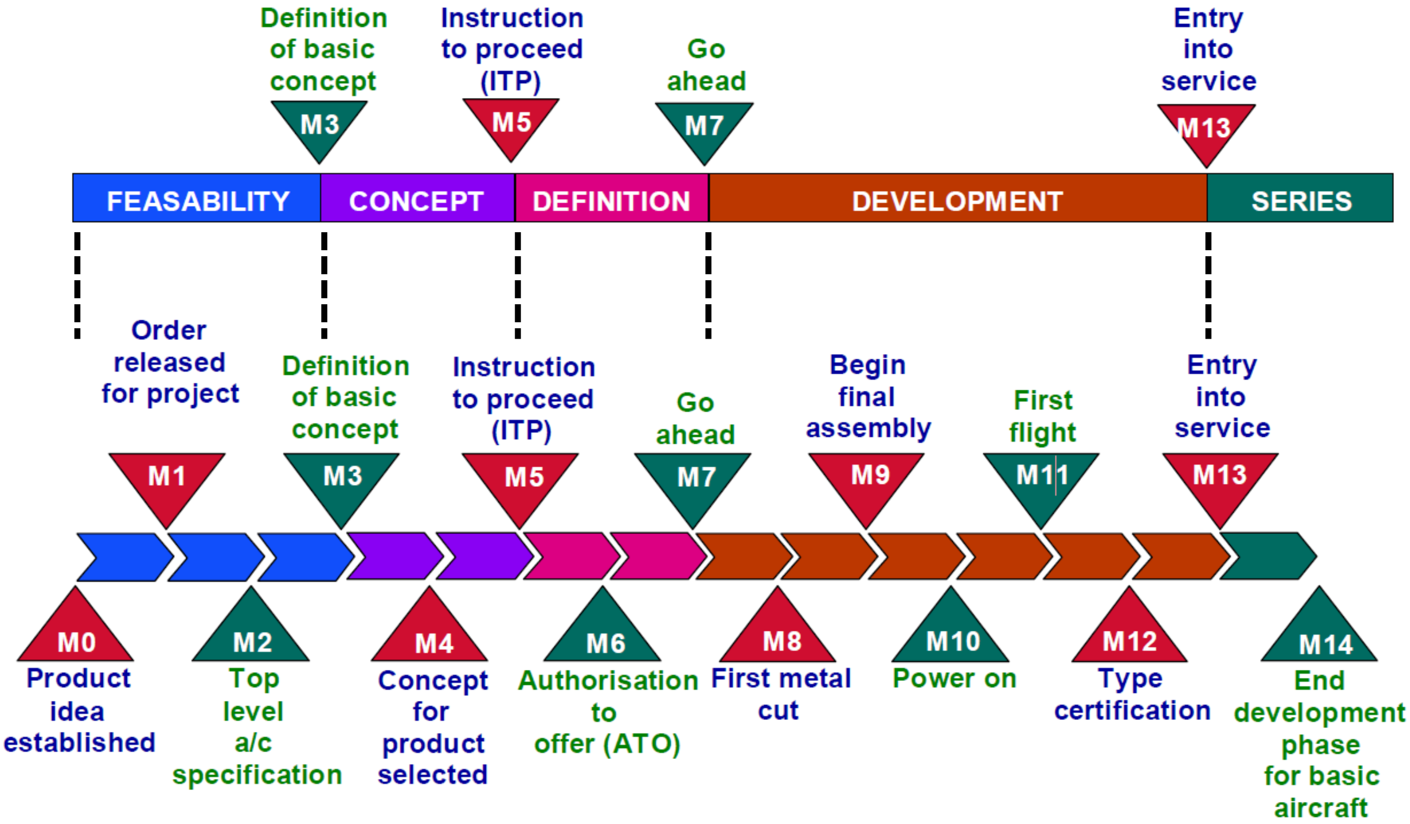
Not one size fits all…

- Type of technology
- Company methods (and culture)
- New innovation or new application
- Customer (and supplier systems)

Project Lifecycle(s)



Example: Airbus



[1] Pardessus T., 2004, Concurrent engineering development and practices for aircraft design at Airbus, 24th International Congress of the Aeronautical Sciences (ICAS), available at http://www.icas.org/ICAS_ARCHIVE/ICAS2004/PAPERS/413.PDF

Diversity of projects

Development of a new landing gear concept

Development of a new cloud based tool for designing and purchasing home interiors

As a team - discuss how these projects might differ. Use the table on the right to prompt you.

- ❑ Safety criticality
- ❑ Duration of tasks
- ❑ Time to build the prototype
- ❑ Interdependency of tasks (how sequential?)
- ❑ Legislation
- ❑ How iterative can the design/project/manufacture be?
- ❑ Quality documentation for legislator and/or customer
- ❑ Testing and validation
- ❑ Clearly defined end goal and/or customer requirements

Waterfall vs Agile

Prototype 1

Field Trial

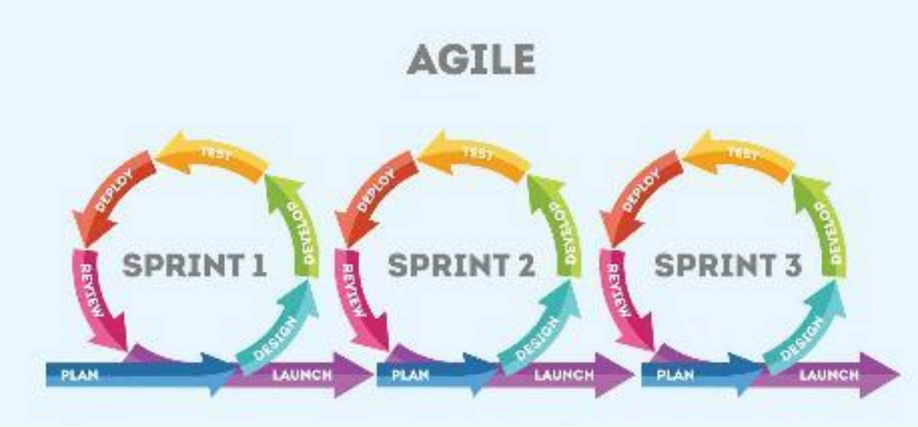
Design

Manufacture

Test

Validate

Design



https://www.soldevelo.com/

Very Sequential
Dependent tasks
Predictable
Regulated
Clear requirements
Lots of documentation
Expensive and time-consuming prototypes

End goal likely to change
Less documentation
Unpredictable / Iterative
Better to get something that works partially
Scrum/Kanban

More than waterfall vs Agile

Producing a project plan… tools

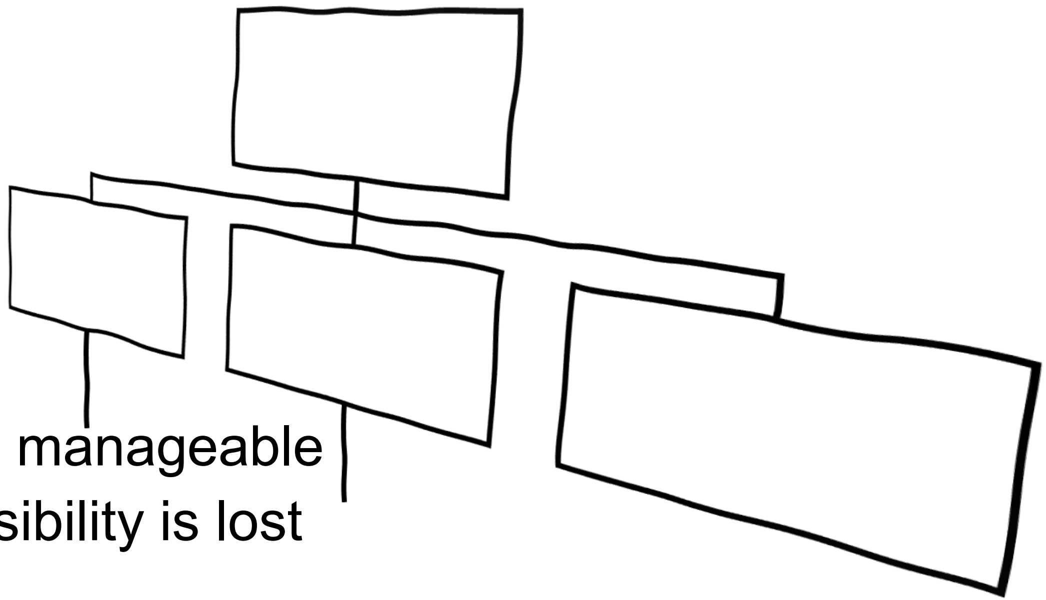
Work Breakdown Structure	- Breaks down the project into a manageable set of tasks - Arranges the tasks in a logical way
Network Diagram	Determine duration of tasks and relationships between them
Gantt	Plot tasks and relationships in-time

Project scheduling tools

Work breakdown structure

1) Breaking down large projects into smaller* tasks

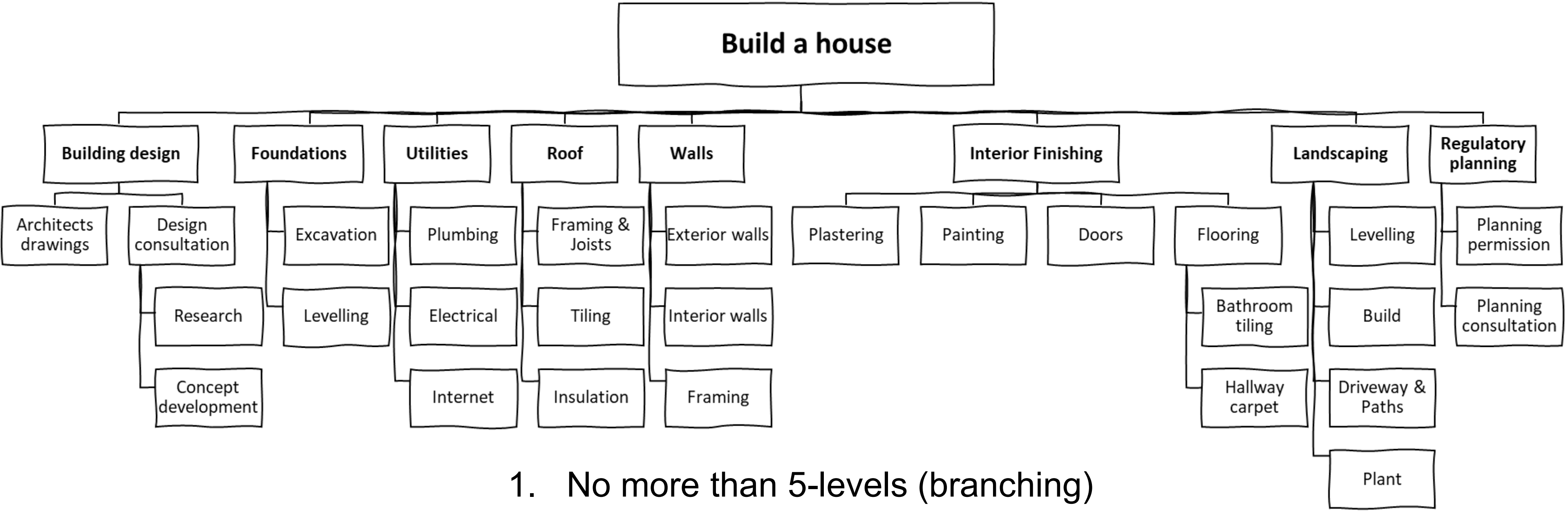
2) Then arrange logically



*Small enough to be manageable

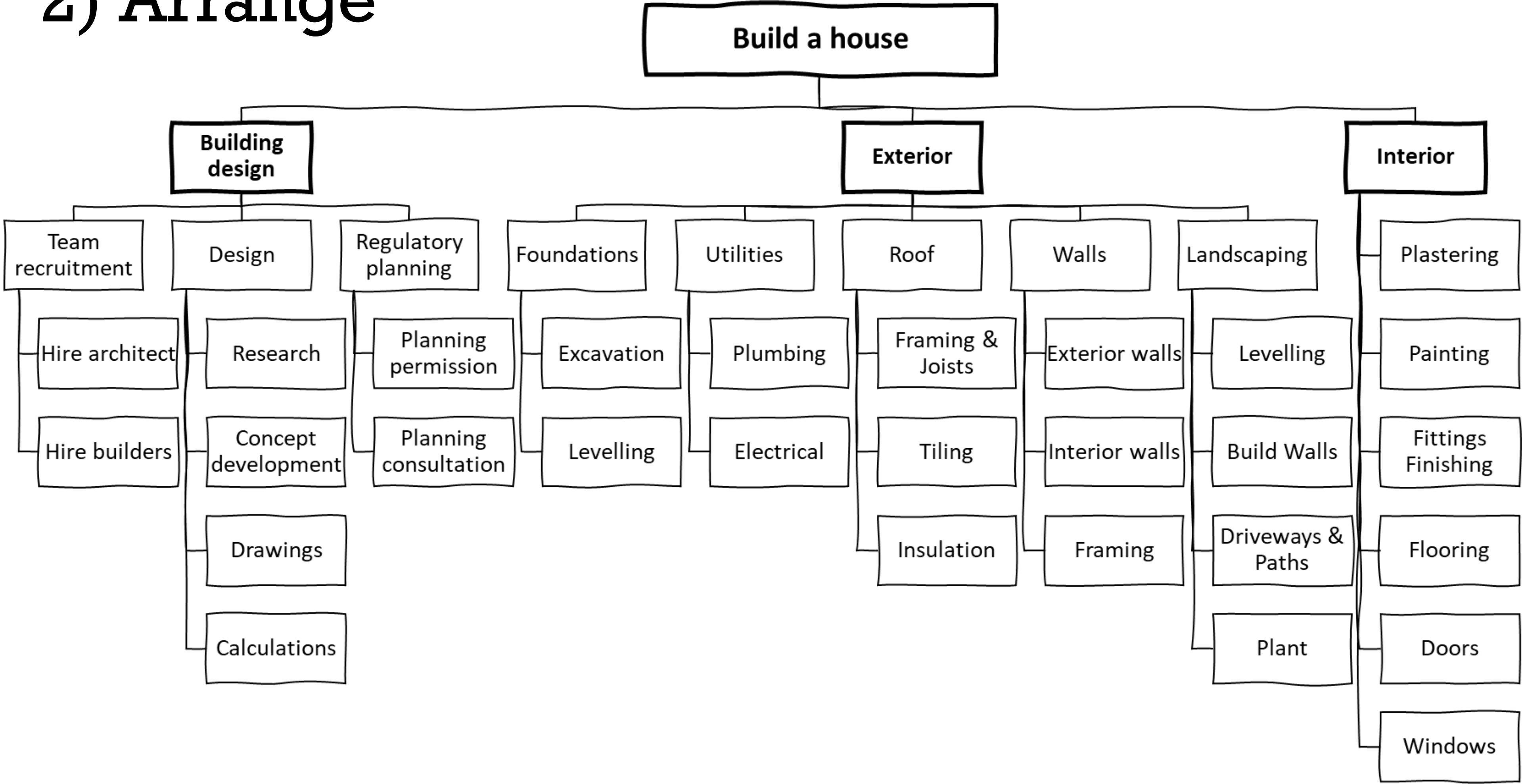
*Not so small that visibility is lost

1) Breakdown



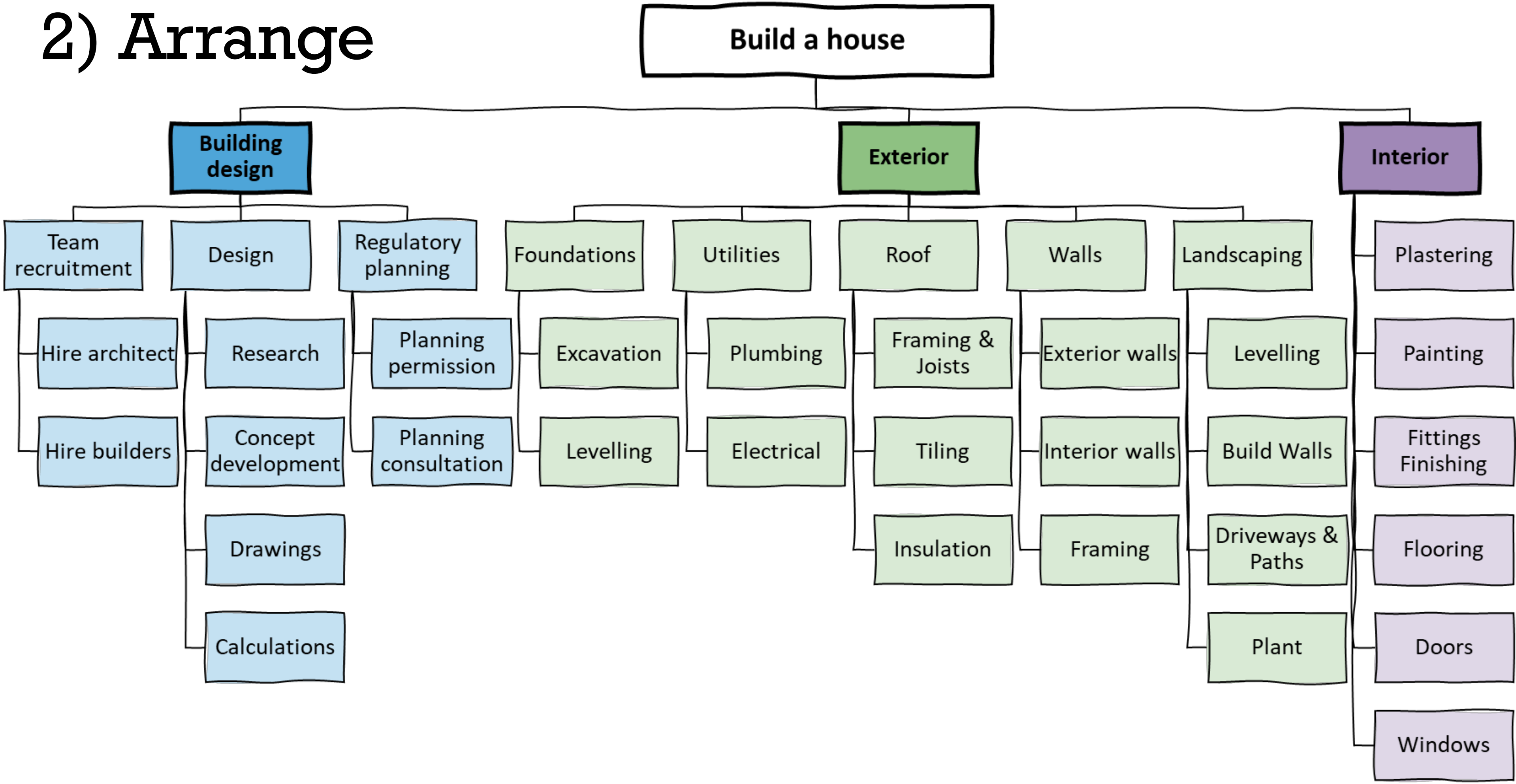
1. No more than 5-levels (branching)
2. Captures everything (the 100% rule)
3. Tasks contribute to parent task

2) Arrange



Build a house
Building design
Exterior
Interior
Team recruitment
Design
Regulatory planning
Foundations
Utilities
Roof
Walls
Landscaping
Plastering
Hire architect
Research
Planning permission
Excavation
Plumbing
Framing & Joists
Exterior walls
Levelling
Painting
Hire builders
Concept development
Planning consultation
Levelling
Electrical
Tiling
Interior walls
Build Walls
Fittings Finishing
Drawings
Insulation
Framing
Driveways & Paths
Flooring
Calculations
Plant
Doors
Windows

2) Arrange



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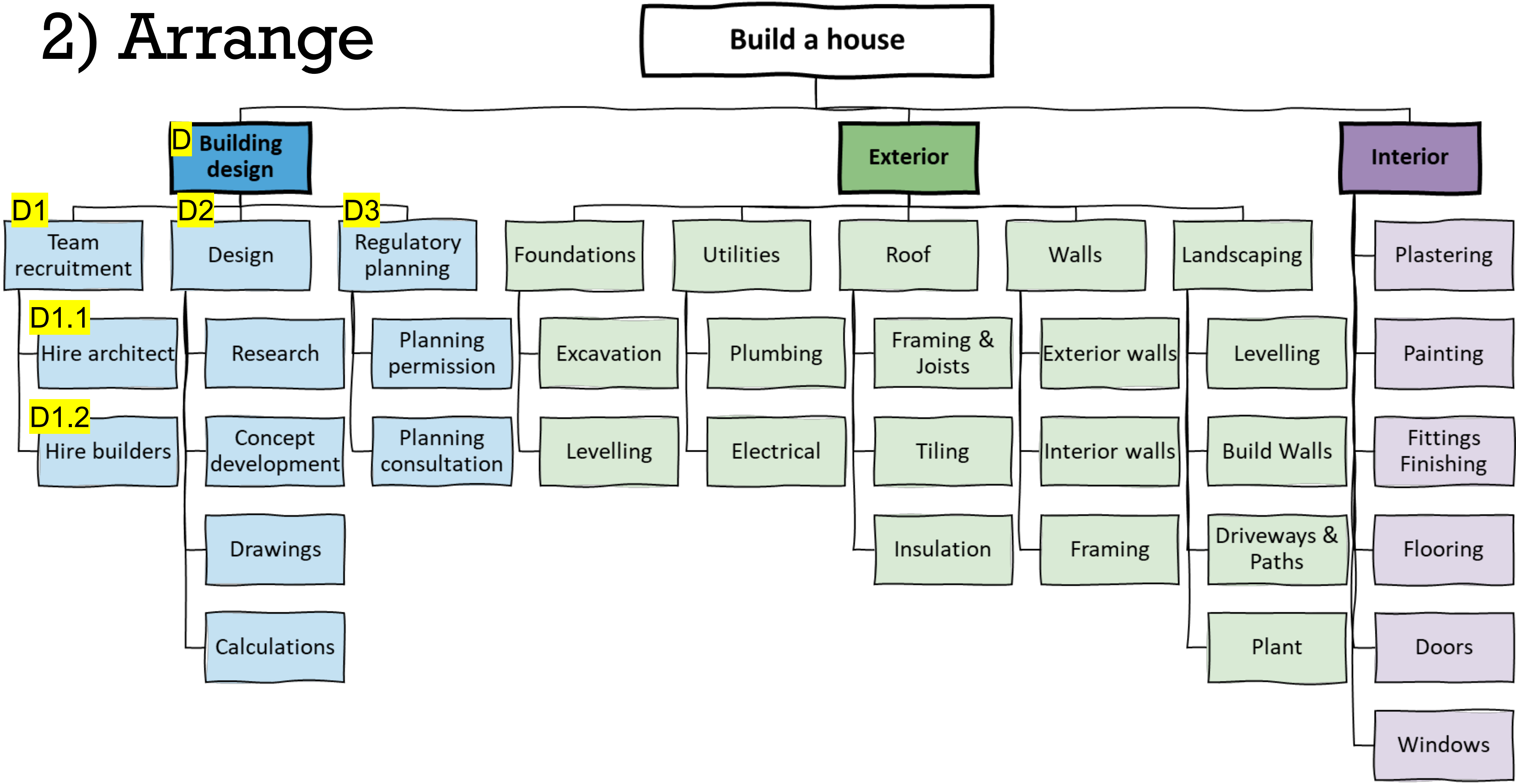
2) Arrange


Build a house
D Building design
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D1 Team recruitment
D2 Design
D3 Regulatory planning
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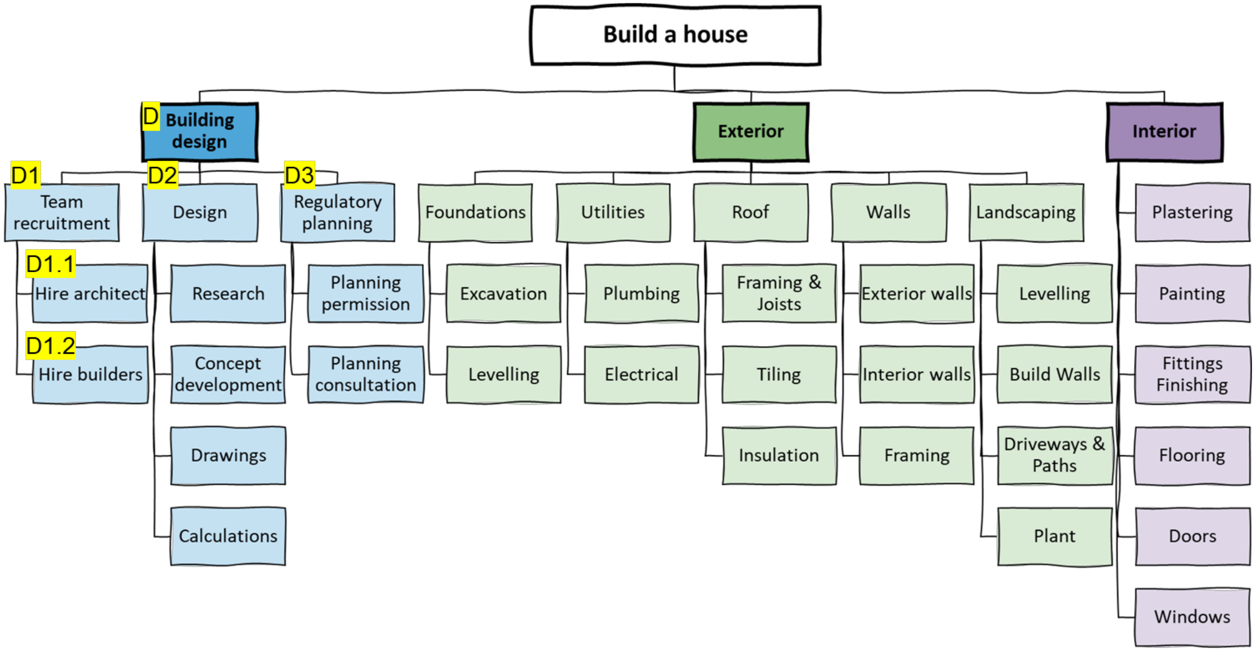
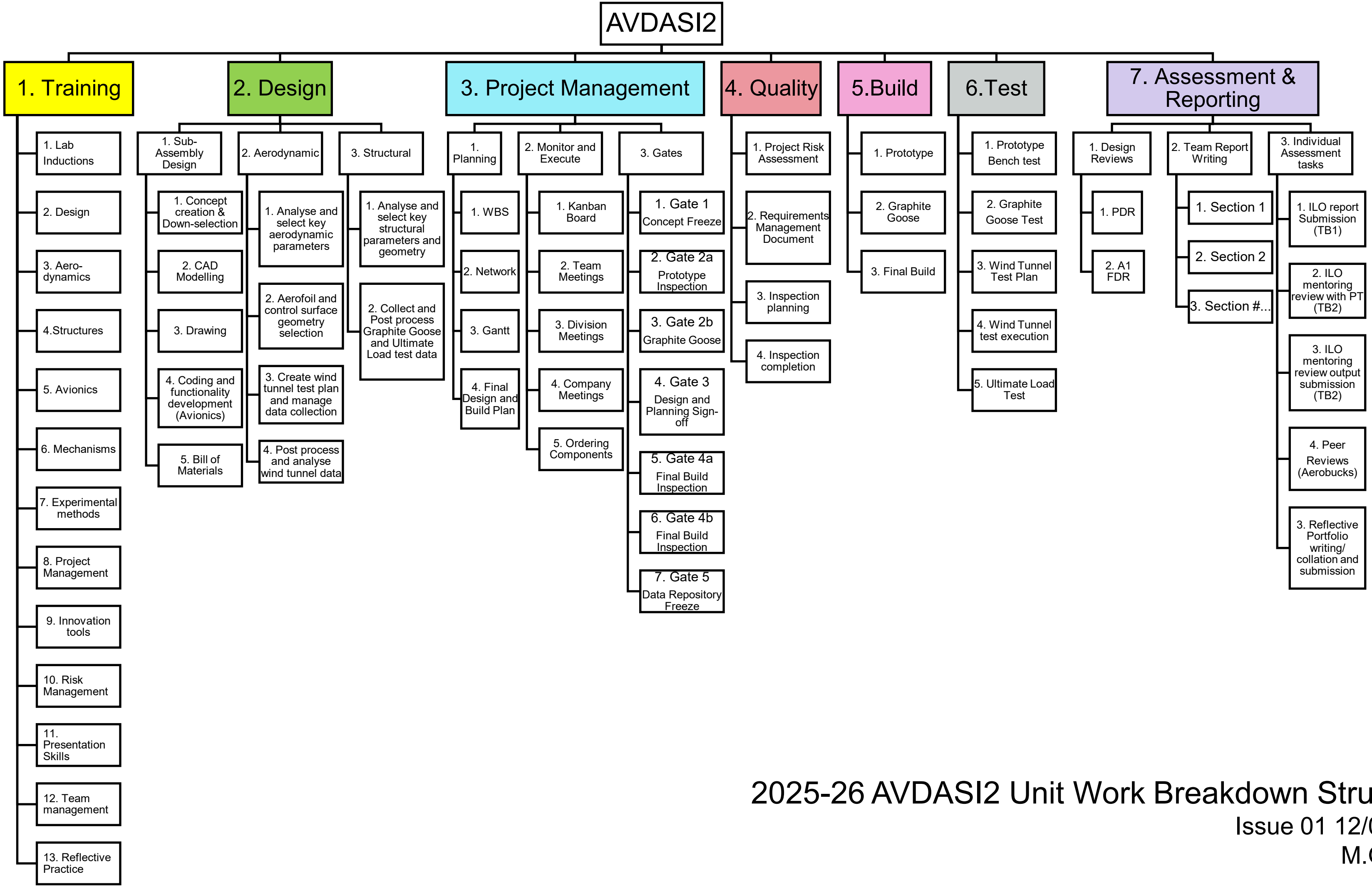
WBS



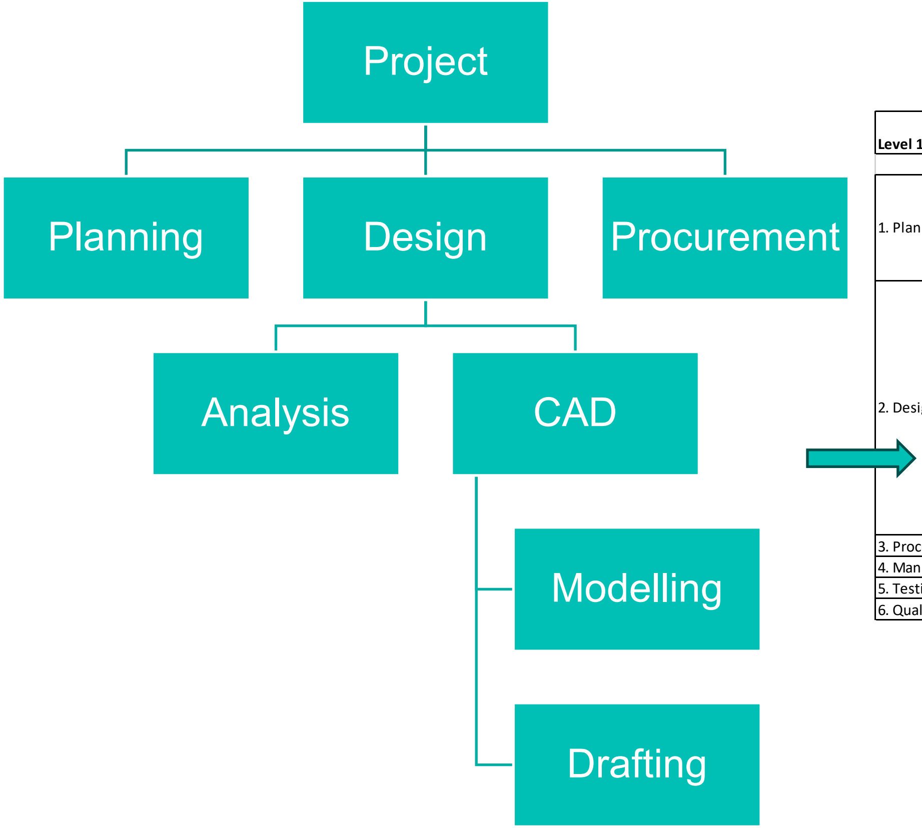
WBS

- ✓ What work needs to be done
- × How long the task takes
- × What the **dependencies** are
- × How long the project takes



2025-26 AVDASI2 Unit Work Breakdown Structure
Issue 01 12/09/2025
M.Graham

WBS > List > Team Kanban Board



Level 1	Level 2	Level 3	Level 4	Work package description	Link to task on Kanban
1. Planning	1.1 WBS	1.1.1 Release first issue of WBS			
	1.2 Network	1.2.1 Create Dependencies Table			
		1.2.2 Create Network Diagram			
	1.3.1 Gantt	1.3.1 Create Gantt Chart			
2. Design	2.1 Analysis	2.1.1 Structural analysis	2.1.1.1 Main Wing Spar structural calculations		
			2.1.1.2 MWP-Emp Joint structural calculations		
		2.1.2 Aerodynamic analysis	2.1.2.1 Main Wing Spar structural calculations		
	2.2 CAD	2.2.1 Modelling	2.2.1.1 Model Main Wing Spar		
			2.2.1.2 Model Wing Ribs and Skin		
		2.2.2 Drafting	2.2.1.1 Release Main Wing Spar (Dwg 0008)	Draw, Check, Approve and release	https://tasks.office.com/bristol.ac.uk/en-
			2.2.1.2 Release Wing Ribs and Skin (Dwg 0015)		
3. Procurement	...	...	...		
4. Manufacturing	...	...	...		
5. Testing	...	...	...		
6. Quality	...	...	...		

WBS > List > Team Kanban Board

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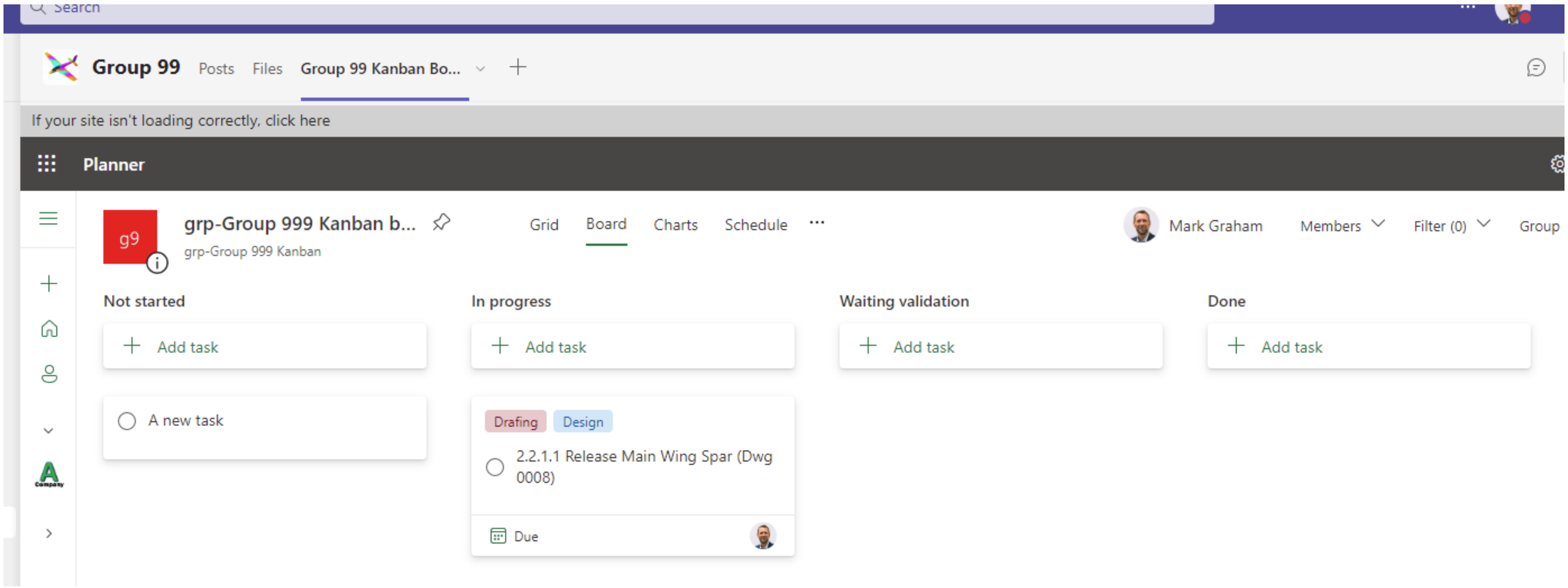
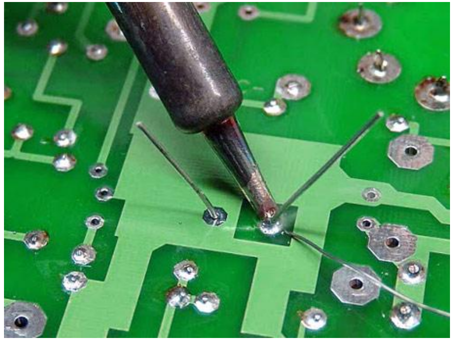
3. Procurement	…	…	…		
4. Manufacturing	…	…	…		
5. Testing	…	…	…		
6. Quality	…	…	…		

Team Kanban board



All communications and task management (for the group) should take place on your private teams channel.

Network diagram (activity-on-arrow)



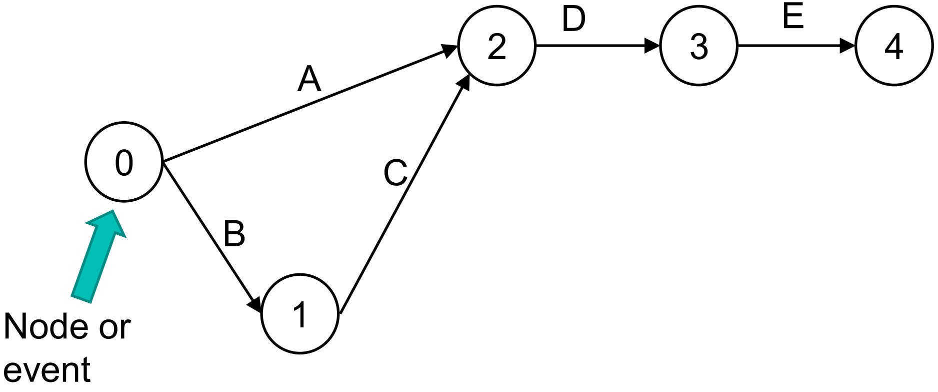
Activity	Description	Predecessor
A	?	(Start)
B	?	(Start)
C	?	?
D	?	?
E	?	?

Activities from WBS

- Solder Parts
- Collect all parts from store
- Fit parts to PCB (Printed circuit board)
- Test PCB
- Switch on soldering iron

Network diagram (activity-on-arrow)

Activity	Description	Predecessor
A	Warm-up iron	(Start)
B	Collect all parts	(Start)
C	Fit parts to PCB	B
D	Solder parts	A,C
E	Test	D

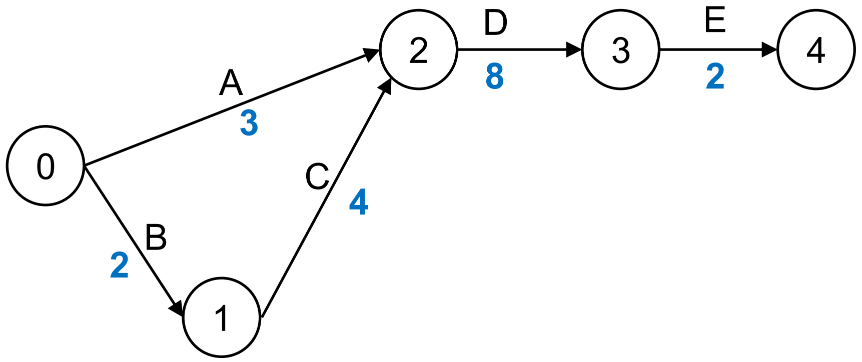


Rules:
- Time: left to right
- Nodes = Events (start/end)
- There is no scale
- Arrows = Activities/Tasks
- Connections = dependencies

Layout the network

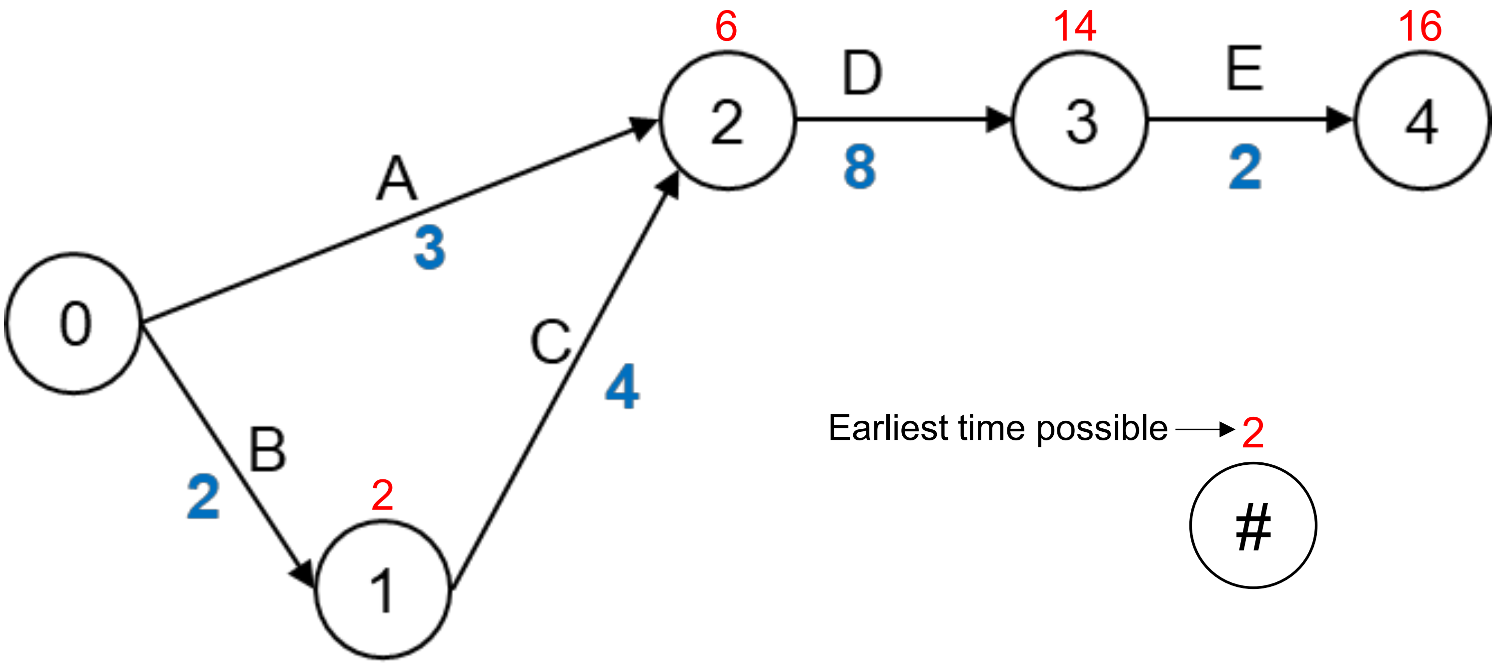
Network diagram (activity-on-arrow)

Activity	Description	Predecessor	Duration (mins)
A	Warm-up iron		3
B	Collect all parts		2
C	Fit parts to PCB	B	4
D	Solder parts	A,C	8
E	Test	D	2



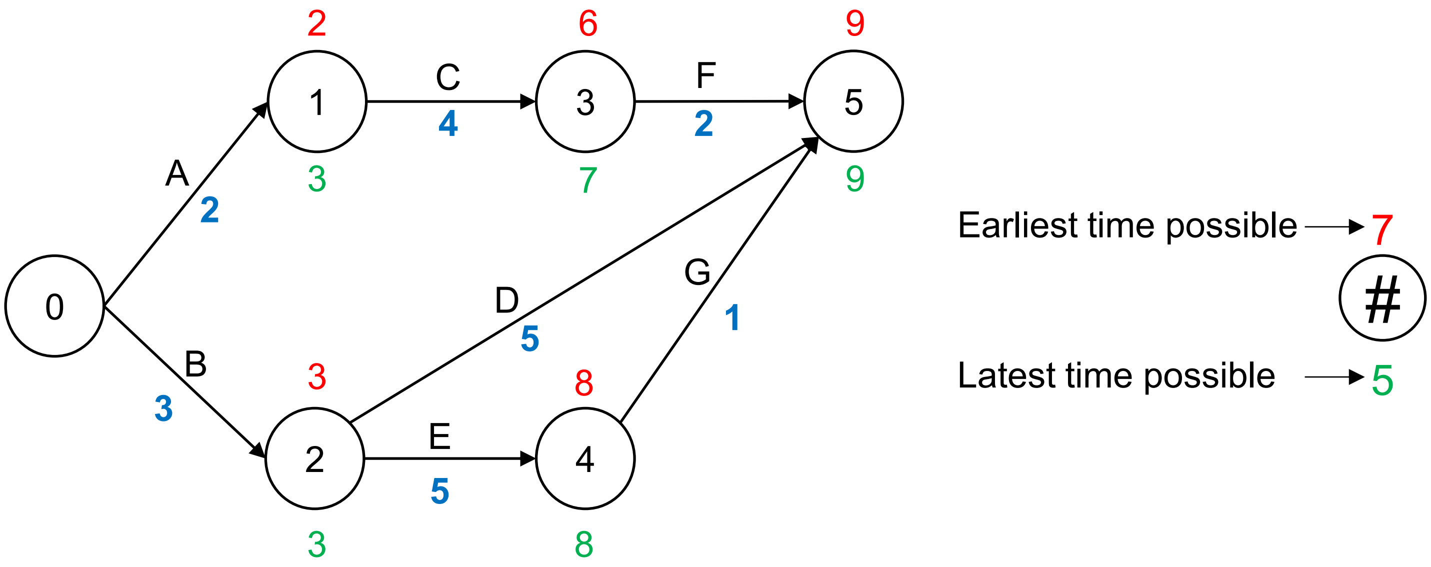
Forward pass planning…. Add durations

Forward pass planning



Forward pass planning…. calculate earliest start

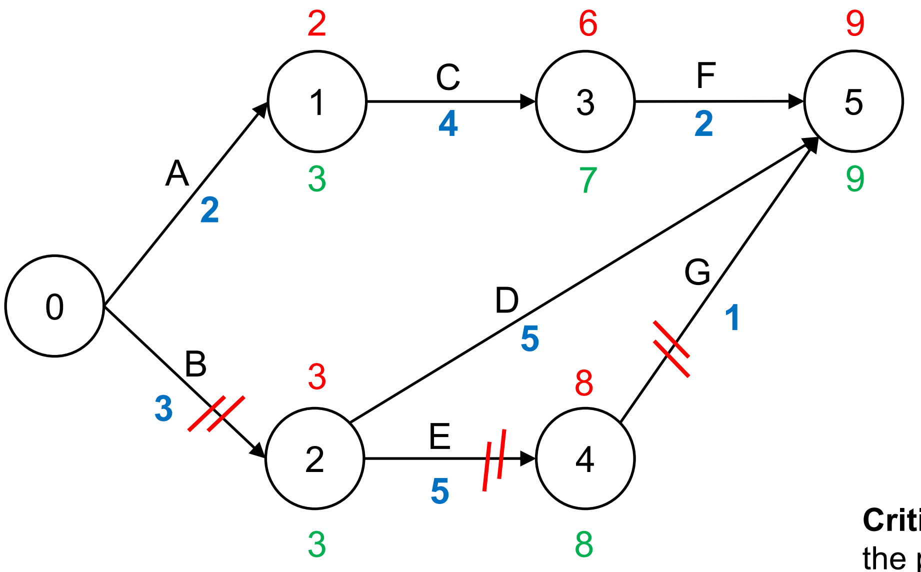
Forward pass and backward pass



Forwards pass…. Calculate earliest start

Backwards pass… Calculate latest finish

Forward pass and backward pass



Earliest time possible → 7

#

Latest time possible → 16

Critical path = the longest path though the project

Critical path = 9 days

Critical path = B → E → G

Backwards pass… Critical path

Your turn

Activity	Predecessor	Duration (days)
A	(Start)	3
B	(Start)	2
C	A	4
D	B	5
E	B	4
F	C,D	3
G	E	2

Instructions

1. Draw the network
2. Label the durations
3. Conduct forward pass (identify earliest possible time at each node)
4. Conduct the backwards path
5. State the critical path length
6. Identify and state the critical path(s) route(s) i.e. the activity sequence(s)
7. **Bonus:** Which task(s) can be delayed (and by how much) without delaying the project?

Rules & conventions

Activities drawn as arrows

Events (nodes) as circles

Time increases left to right

Label the critical path with //

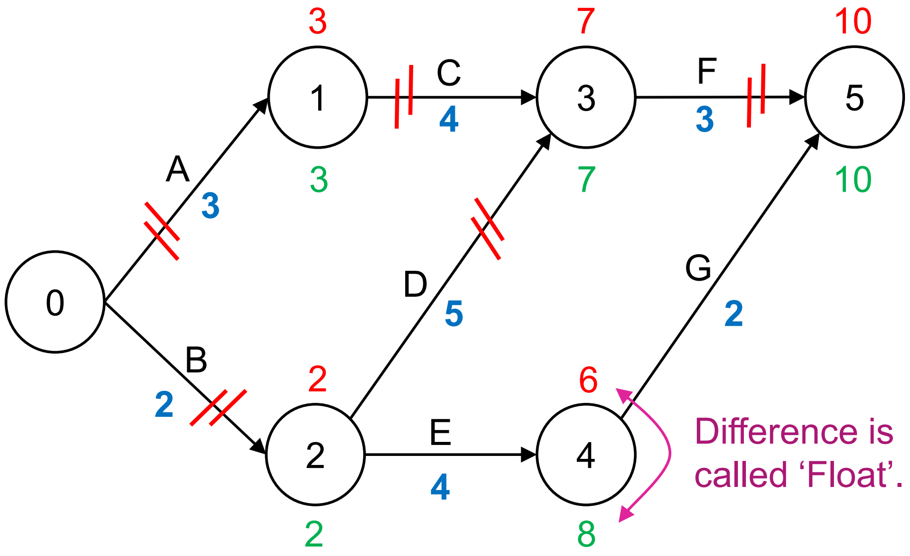
Show durations below the activity letter

Start = Node 0

Record earliest finish above and latest start below

Your turn

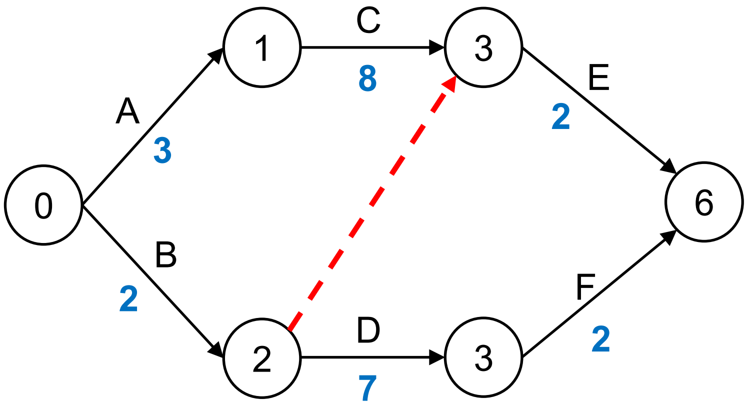
Activity	Predecessor	Duration (days)
A	(Start)	3
B	(Start)	2
C	A	4
D	B	5
E	B	4
F	C,D	3
G	E	2



Critical path = 10 days
Critical path(s) = A → C → F *and* B → D → F

Tasks E and G can be delayed by a <u>total</u> of 2 days without affecting the project duration…i.e. "there are 2 days of float"

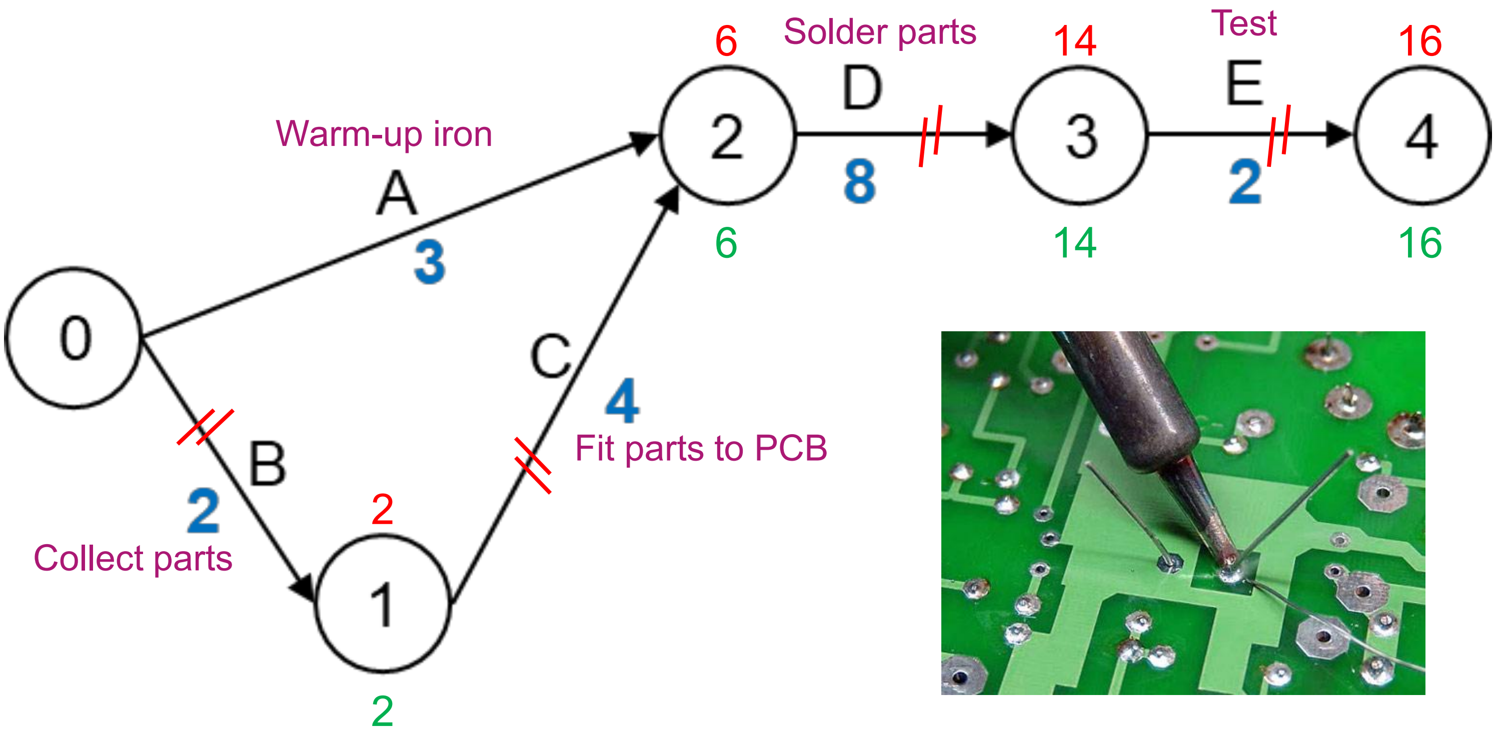
Dummy activities

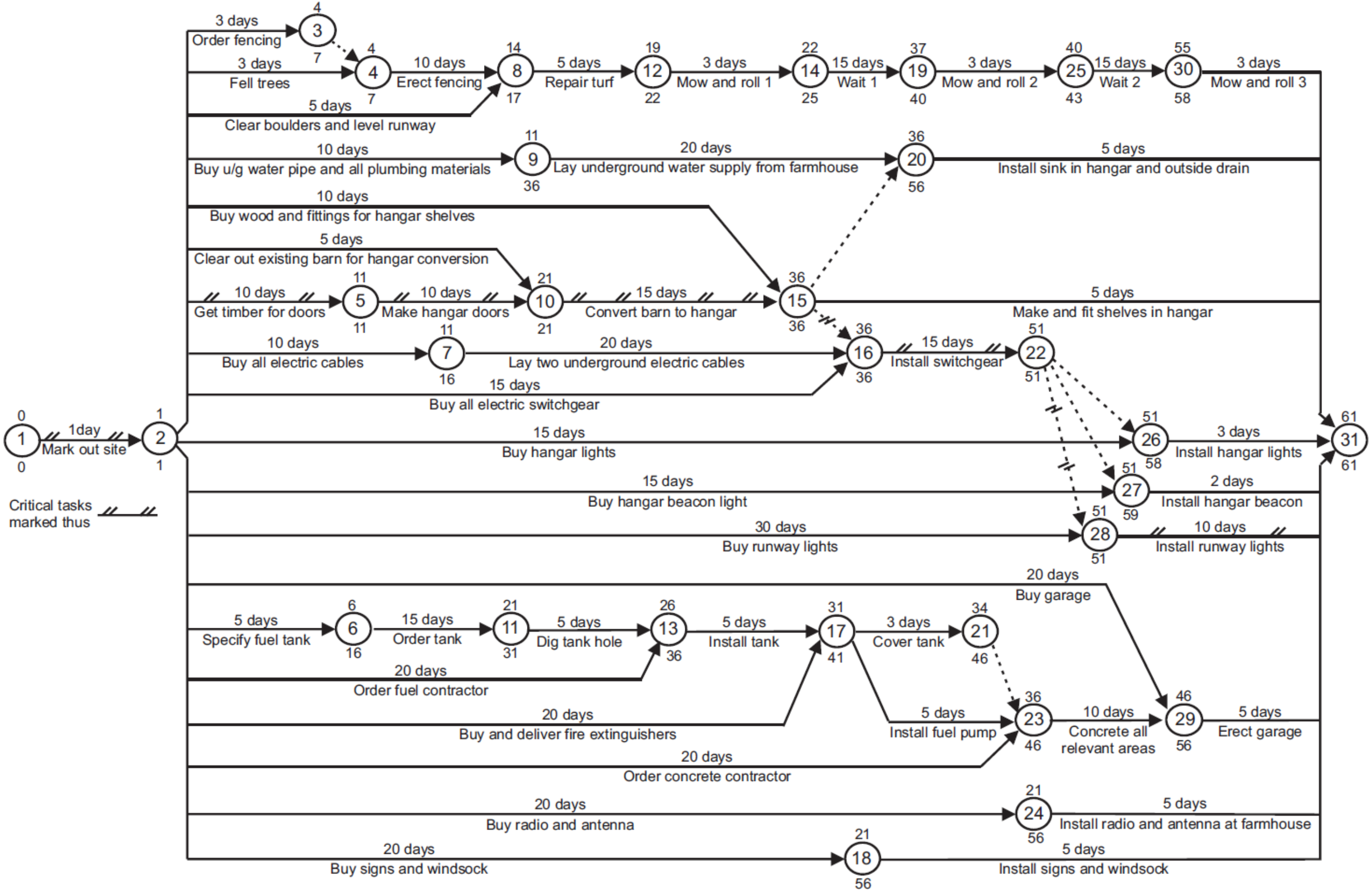


Dummy activity: Representing a dependency, but line does not represent an activity (time)

Activity	Predecessor	Duration (weeks)
A	(start)	3
B	(start)	2
C	A	8
D	B	7
E	C, B	2
F	D	2

Network optimisation (real management)





Order fencing
3 days
Fell trees
3 days
Erect fencing
10 days
Repair turf
5 days
Mow and roll 1
3 days
Wait 1
15 days
Mow and roll 2
3 days
Wait 2
15 days
Mow and roll 3
3 days
Clear boulders and level runway
5 days
Buy u/g water pipe and all plumbing materials
10 days
Lay underground water supply from farmhouse
20 days
Install sink in hangar and outside drain
5 days
Buy wood and fittings for hangar shelves
10 days
Clear out existing barn for hangar conversion
5 days
Get timber for doors
10 days
Make hangar doors
10 days
Convert barn to hangar
15 days
Make and fit shelves in hangar
5 days
Buy all electric cables
10 days
Lay two underground electric cables
20 days
Install switchgear
15 days
Buy all electric switchgear
15 days
Mark out site
1day
Buy hangar lights
15 days
Install hangar lights
3 days
Buy hangar beacon light
15 days
Install hangar beacon
2 days
Buy runway lights
30 days
Install runway lights
10 days
Critical tasks marked thus
Buy garage
20 days
Specify fuel tank
5 days
Order tank
15 days
Dig tank hole
5 days
Install tank
5 days
Cover tank
3 days
Order fuel contractor
20 days
Buy and deliver fire extinguishers
20 days
Install fuel pump
5 days
Concrete all relevant areas
10 days
Erect garage
5 days
Order concrete contractor
20 days
Buy radio and antenna
20 days
Install radio and antenna at farmhouse
5 days
Buy signs and windsock
20 days
Install signs and windsock
5 days

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